

STEP I - 1997

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Abstract

A collection of solutions to STEP questions for my own use and also for former colleagues and friends. These are very much incomplete. I aim to complete some sporadically although I make absolutely zero promise of doing so with any sort of regular schedule. I largely do this for fun. Although I try my best, I have undoubtedly made several errors so please do use with caution. Any errors, please do let me know at ccbarnesni@gmail.com.

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11. The equation of motion is

$$-(mg + kv^2) = m\ddot{x}$$

which may be solved as follows

$$\begin{aligned} -g - \frac{k}{m}v^2 &= \frac{dv}{dt} \\ -\int dt &= \int \frac{1}{g + \frac{kv^2}{m}} dv \\ -gt + c &= \int \frac{1}{1 + \frac{kv^2}{mg}} dv \\ -gt + c &= \sqrt{\frac{mg}{k}} \arctan \left(\sqrt{\frac{k}{mg}} v \right) \end{aligned}$$

Applying the initial condition $v(0) = u$ we have

$$\arctan \left(\sqrt{\frac{ku^2}{mg}} \right) - \arctan \left(\sqrt{\frac{kv^2}{mg}} \right) = \sqrt{\frac{kg}{m}} t$$

Taking tangents of either side and applying the compound angle formula for tan with $\alpha = \sqrt{\frac{k}{mg}}$ we obtain

$$\frac{\alpha u - \alpha v}{1 + \alpha^2 uv} = \tan(\alpha gt)$$

Rearranging for v yields

$$\begin{aligned} v &= \frac{\alpha u - \tan(\alpha gt)}{\alpha(1 + \alpha u \tan(\alpha gt))} \\ v &= \frac{\alpha u \cos(\alpha gt) - \sin(\alpha gt)}{\alpha(\cos(\alpha gt) + \alpha u \sin(\alpha gt))} \end{aligned}$$

which is integrated to obtain

$$s(t) = \frac{1}{\alpha^2 g} \ln |\cos(\alpha g t) + \alpha u \sin(\alpha g t)|$$

Thus we have

$$v^2 = \frac{1}{\alpha^2} \cdot \frac{1}{\alpha^2 u^2 \sin^2}$$

12.

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16.